

MOHIT BHALKE

Associate Software Engineer

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Data-driven professional skilled in data analysis, machine learning, and generative AI. Experienced in building predictive models, optimizing workflows, and delivering actionable insights. Proficient in Python, SQL, and frameworks such as Scikit-learn, LangChain, Langgraphs, and Flask, with hands-on experience developing scalable APIs and intelligent solutions that drive measurable business impact.

Experience

Associate Software Engineer | DeltaFx Technologies

May 2023 - Present

- Designed and implemented **interactive dashboards** to visualize key metrics for business performance and client trading behavior.
- Built a Generative AI-based chatbot using **Hugging Face Transformers**, leveraging internal CRM and support documents by **embedding** user queries, performing **vector similarity search**, and generating **contextual responses** with an LLM to automate common client query resolution.
- Developed and integrated an NLP-based **sentiment analysis** tool into the CRM system, triggering email alerts only when negative sentiment exceeds a defined threshold, streamlining critical escalation workflows.
- Took initiative to **optimize existing codebase**, reducing code redundancy by **25%**, increasing modularity, and enhancing system maintainability.

Data Analyst | Careervira

July 2022 - July 2023

- Collaborated with cross-functional teams (**marketing, product, and tech**) to define data requirements and deliver actionable insights for strategic decision-making.
- Performed **data extraction, cleaning, and analysis** on user behavior and course engagement data using **Python (Pandas, NumPy), SQL, and Excel** and Automated repetitive workflows using Python scripts and **Google AppScript**, cutting down recurring report generation time by **50%**
- Conducted **statistical analysis** and **A/B testing** to evaluate marketing strategies and feature effectiveness.
- Developed **predictive models** for **user segmentation** and **sentiment analysis**, boosting the efficiency of personalized marketing by up to **35%**.

Data Science Intern | YoShops

April 2022 - June 2022

- Collaborated with a diverse team of interns spanning various departments, contributing to projects involving the development of **machine learning models, web scraping techniques, and sentiment analysis**.
- Simultaneously participated in an intensive 8-week training program in **Data Science & Machine Learning**.

Skills

- Programming Languages:** Python, PHP
- Machine Learning & Generative AI:** Supervised & Unsupervised Learning, NLP, Predictive Modeling, LLMs, PyTorch, LangChain, LangGraph, RAG, Hugging Face Transformers, OpenAI API
- Data Science & Analytics:** Statistical Analysis, Data Cleaning, Feature Engineering, Predictive Analytics, ETL, Web Scraping (Scrapy, Selenium), Excel
- Frameworks & APIs:** Django, Flask, FastAPI, CodeIgniter3, REST APIs
- Databases & Cloud:** MySQL, MongoDB, Elasticsearch, Vector Databases (e.g., Pinecone, FAISS), AWS
- Tools & Libraries:** Scikit-learn, Pydantic, Pandas, NumPy, NLTK
- Data Visualization:** Matplotlib, Seaborn, Streamlit, Dash

Projects

● Recommendation System

Stack: Python, Streamlit, HuggingFace, LangChain, Pinecone, OMDb API

- Built a GenAI-powered **movie recommendation system** using **BERT-based sentence embeddings** and **Pinecone vector DB**, enabling semantically rich and personalized movie suggestions.
- Transitioned from traditional NLP (TF-IDF + **cosine similarity**) to a modern GenAI stack (**HuggingFace** + **LangChain**), improving contextual relevance and scalability for large datasets.
- Deployed an interactive **Streamlit** web app with poster previews, detailed IMDb metadata, and a "You may also like" section powered by **vector-based similarity retrieval**.

● Commodity Price Predictor

Stack: Python, Streamlit, Scikit-learn, XGBoost, GridSearchCV, Joblib, Pandas

- Designed and deployed a unified machine learning solution for predicting prices across **cars, mobile phones, and houses** using structured datasets and domain-specific **regression models**.
- Built robust preprocessing pipelines with **OneHotEncoding**, **target encoding**, and **feature scaling**, enabling consistent handling of **categorical and numerical data** across commodity types.
- Achieved strong predictive performance, achieving optimal **R² scores up to 91%**, with **MAE and RMSE metrics** tailored per domain for transparency and model effectiveness.
- Improved model reliability and accuracy using **GridSearchCV-based hyperparameter tuning**, minimizing error margins across all three commodities.
- Designed a modular **Streamlit** application featuring a dynamic UI that adapts input fields based on selected commodity, allowing seamless interaction and **real-time predictions**.

● Sentiment Escalator

Stack: Python, FastAPI, HuggingFace (RoBERTa), REST API, Pydantic, Logging

- Built a **FastAPI-based sentiment analysis** microservice providing real-time sentiment scores via secure **REST endpoints**.
- Leveraged **HuggingFace's RoBERTa model (cardiffnlp/twitter-roberta-base-sentiment)** for robust text classification in customer support tickets.
- Added **rotating logs, error handling**, and health-check endpoints to ensure production reliability and observability.
- Positioned for future enhancement with fine-tuning on domain-specific datasets and hyperparameter tuning for improved accuracy.

Education

● B-Tech Honor's | Data Science | Sanjivani College Of Engineering

● B-Tech | Mechanical Engineering | Sanjivani College Of Engineering

● HSC | Science | New English Jr. College

● SSC | St. Xavier's High School

Certification

- Applied Machine Learning in Python | Coursera
- Statistics for Data Science with Python | Coursera
- Fundamental Of Visualization with Power Bi | Coursera
- Deep Learning | NPTEL
- SQL For Data Science | Coursera
- AWS Academy Cloud Foundations | AWS
- Visual Analytics With Tableau | Coursera
- Python Programming | NPTEL

Language

- English
- Hindi
- Marathi